Hearing & Perception

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Hearing vs Perception

Hearing

Physical and anatomical processes in the ear

Perception

Sense making of what was heard

Hearing vs Perception

What you *hear* ≠ what you *percieve*

Anatomy

Outer Ear

- pinna
- ear canal
- tympanic membrane

Anatomy

Middle Ear

- malleus
- incus
- stapes

Anatomy

Inner Ear

- Cochlea

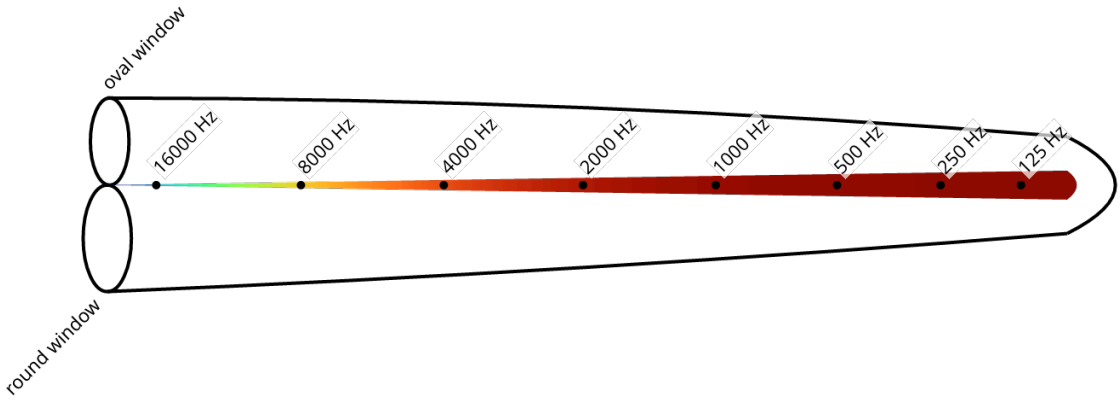
Cochlea

- The shortest possible refractory period of a nerve firing is about 0.5 ms

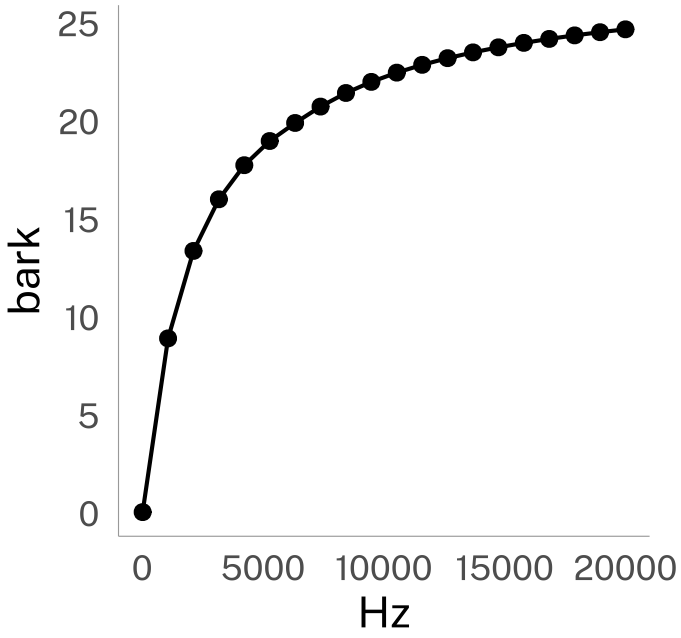
- That means the highest frequency a nerve could *possibly* send signals at is 2000 Hz

Cochlea

Cochlea (unfurled)



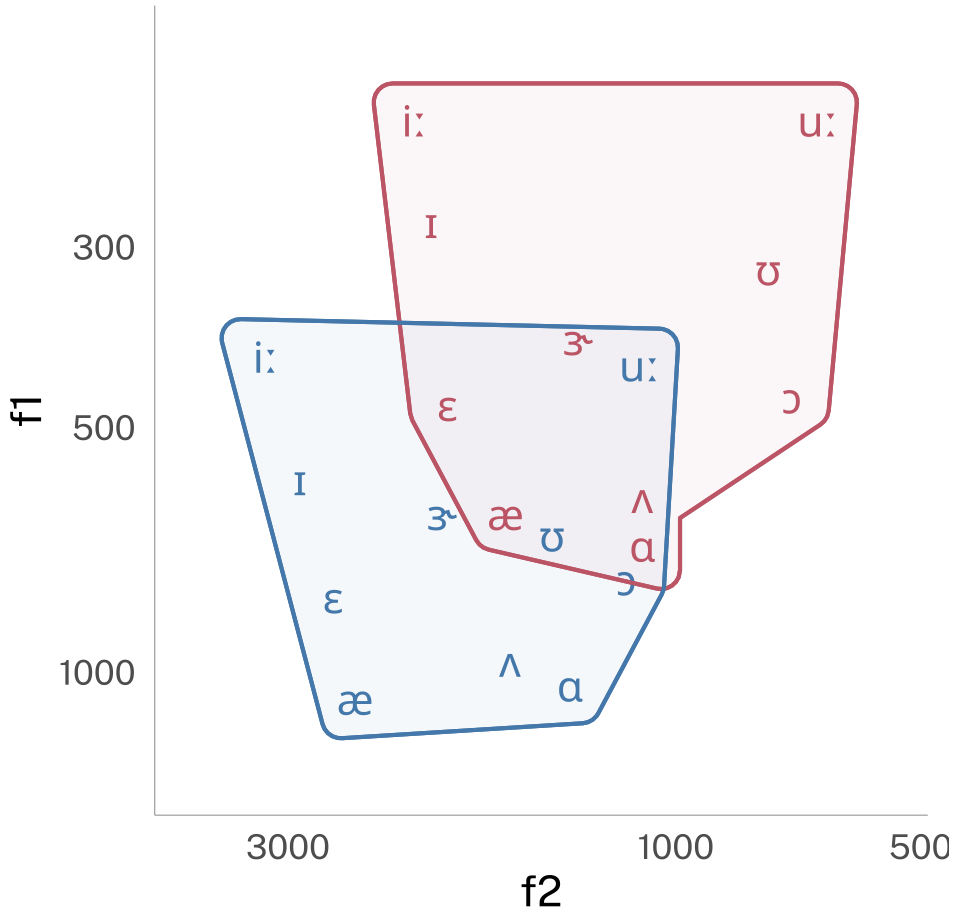
Perception - Nonlinear



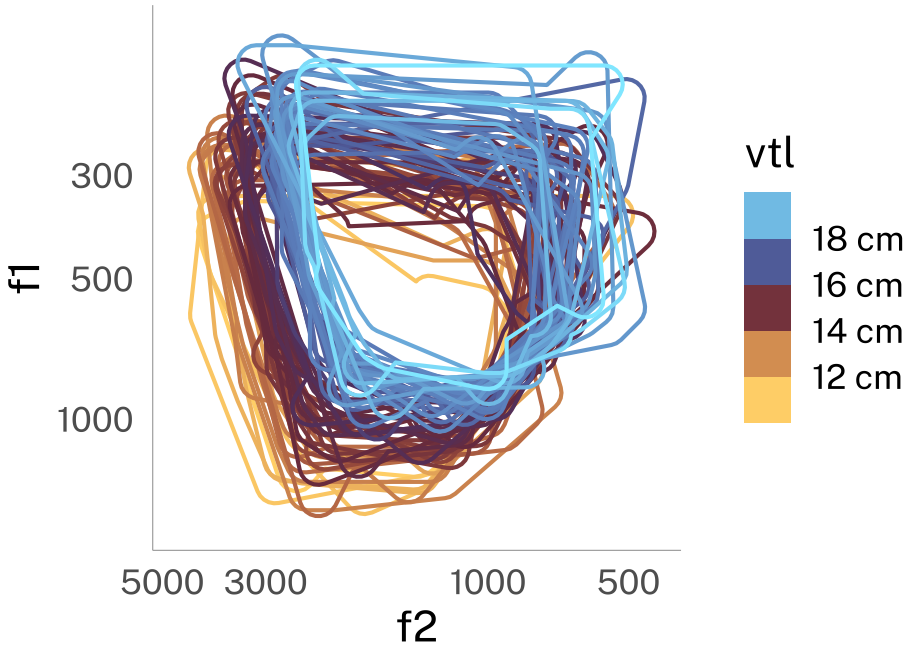
Perception - The Brain

We do not understand exactly how the brain creates the perception of language. (Zsiga, 2024).

Perception - Speech Variability



Perception - Speech Variability



Perception - Cue Integration

https://www.youtube.com/watch?v=aFPtc8BVdJk

Perception - Predictive

```
viewof spect2 = {
  const img = html`<img src="./images/silent_${step2}.svg"></img>`
  return img;
}
viewof myAudio2 = {
     const audio = html`<audio controls src="./silent_center/silences/
clip_${step2}.wav"></audio>`;
  // Expose the audio element's state or actions reactively
  return audio;
}
viewof step2 = Inputs.range([1, 10], {value: 10, step: 1, label: "step"})
```



Perception - Categorical (?)

```
viewof spect = {
  const img = html`<img src="./images/clip_${step}.svg"></img>`
  return img;
}
viewof myAudio = {
  const audio = html`<audio controls src="./audio/clip_${step}.wav"></audio>`;
  // Expose the audio element's state or actions reactively
  return audio;
}
viewof step = Inputs.range([1, 20], {value: 20, step: 1, label: "step"})
```